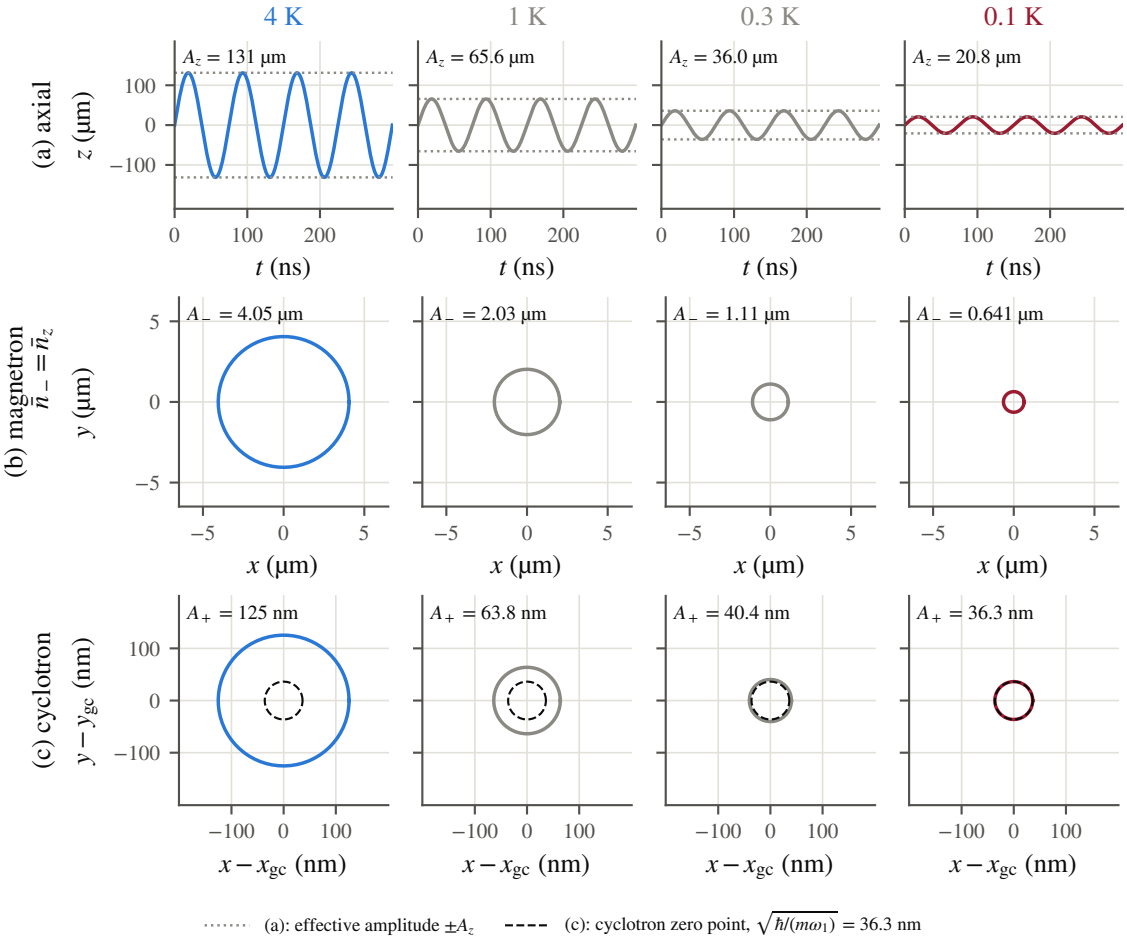
4 K
1 K
0.3 K
0.1 K
(a) axial
z (μm)
A_z = 131 μm
A_z = 65.6 μm
A_z = 36.0 μm
A_z = 20.8 μm
100
0
−100
0
100
200
t (ns)
(b) magnetron
$\bar{n}_- = \bar{n}_z$
y (μm)
A_- = 4.05 μm
A_- = 2.03 μm
A_- = 1.11 μm
A_- = 0.641 μm
5
0
−5
x (μm)
(c) cyclotron
$y - y_{gc}$ (nm)
A_+ = 125 nm
A_+ = 63.8 nm
A_+ = 40.4 nm
A_+ = 36.3 nm
$x - x_{gc}$ (nm)
(a): effective amplitude $\pm A_z$
(c): cyclotron zero point, $\sqrt{\hbar/(m\omega_1)}$ = 36.3 nm